

A Spectrum Sensor for CubeSat Radios

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Abstract

Small-spacecraft missions increasingly use distributed architectures, making awareness of shared radio-frequency spectrum useful for interference mitigation and spectrum sharing. Conventional spectral-correlation analyzers, however, can demand more memory and computation than some CubeSat radios can readily provide. This paper develops a low-memory spectrum sensor for field-programmable gate arrays based on a streaming spectral-correlation analyzer. Under the evaluated signal model, the sensor searches configured symbol-rate and center-frequency candidates for power-of-two-order quadrature amplitude modulation waveforms with square-root-raised-cosine pulse shaping. A power-normalized binary threshold nominates symbol-rate candidates; a center-frequency detector retains the strongest frequency candidate, and a proposed scheduler coordinates parallel detector instances. Monte Carlo simulations with two spectrally overlapping signals were used to characterize the detector. Across the nine reported scenarios, the symbol-rate detection proportion ranged from 0.89 to 1.0, while the false-alarm proportion per unoccupied symbol-rate candidate ranged from 0.0013 to 0.1213. These metrics assess symbol-rate detection only: the archived scoring code does not test center-frequency correctness. Separate returned center-frequency estimates appeared more dispersed as the nominal energy-per-bit-to-noise-density setting decreased. The results support feasibility within the simulated signal conditions, but they do not establish general channel independence, hardware performance, or center-frequency accuracy. A mixed-date engineering archive contains later register-transfer-level source and partial on-device traces, but not an exact reproducible 2021 build or implementation reports.

Index Terms

CubeSat, spectrum sensing, spectral correlation analyzer, field-programmable gate array